

Software Requirements Specification

Worked Example — StudyMate (Instructor's Teaching Demo)

Introduction to Software Engineering (15031001) · Week 3 · Requirements & Specification

Do NOT copy this domain for your own M2. Copy the SHAPE: numbered FRs written as testable user stories, measurable NFRs, an explicit out-of-scope list, one filled Golden Thread row, and an honest AI usage log. Your own SRS is about your own team's problem, not StudyMate.

Field	This example
Team name	MedMind
Project title	Medication Reminder — medication-session reminders and caregiver alerts for older adults
SDLC chosen in M1 + reason	Agile — OCR, notifications, and usability for older adults are uncertain, so we test in short cycles with user feedback
Date / version	v1.0 · 24 September 2026
GitHub repo	https://github.com/IDK-SleepyBro/MedMind

1. Introduction

1.1 Purpose and scope

Older adults who take medication on a schedule may forget a dose or be unsure whether they already took it, and the family members who help them often have no way to know the status of a dose without calling. Setting up medicines and times in a reminder app can also take many steps for people who are not comfortable with smartphones. Medication Reminder lets a caregiver set up the older adult's medicines by photographing the label (reviewed before saving), reminds the older adult at each medication session, lets them confirm the whole session with one tap, and tells the caregiver when a session is still not confirmed after repeated reminders. The MVP is five stories: pairing, label scan, session reminder, one-tap confirm, and escalation. Success looks like this: a caregiver no longer has to call to check every dose, and is alerted by the system when the older adult has not confirmed taking their medicine after the set number of reminders.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
Older adult on scheduled medication	primary user	A clear reminder at the right time and a very simple way to say "I took them"	Complicated screens, typing, passwords; feeling watched or nagged
Family member / caregiver	primary user	Easy setup of medicines and a signal when a dose session is not confirmed	False alarms; wrong medicine details being saved; missing a real problem
Pharmacist / hospital (source of labels)	secondary stakeholder	Label instructions reproduced exactly, never changed	An app that "reinterprets" or changes dosage instructions

2. Overall description

2.1 Product context — what it does / does not

Medication Reminder links one older adult's phone with one or more caregivers' phones, turns caregiver-confirmed medicines into medication sessions, and tracks whether each session is confirmed. Everything below is the whole MVP for this term; nothing else is implied.

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
• FR-1 Pairing (QR or invite code) • FR-2 Label scan → reviewed draft (AI) • FR-3 Session reminder • FR-4 One-tap confirm • FR-5 Escalation to caregiver	• Dosage advice or changing any prescription • Reading handwritten labels • Verifying that medicine was actually swallowed • Full-screen alarm that overrides the phone • iOS app, payments, hospital system integration

2.2 Assumptions and constraints

- Assumption: the older adult uses an Android smartphone (Android 13 or later) and allows notifications; the caregiver has a smartphone.
- Assumption: most labels are printed pharmacy or hospital stickers. [Verify with more label samples]
- Constraint: one Gemini API call per label scan; no model training or fine-tuning.
- Constraint: Gemini free tier may use submitted data, so only de-identified test images are sent during development; no real user label data is sent through the free tier.
- Constraint: health data is sensitive personal data under Thai PDPA; we store only the data listed in FR-1 to FR-5 and need explicit consent before storing it.
- Constraint: Android 14+ restricts full-screen alerts and exact alarms; reminders use standard high-priority notifications.
- Constraint: the M3 prototype is a clickable browser prototype with no real backend; caregiver alerts and timing are simulated there.
- Constraint: Thai and English UI text; no paid services.
- Assumption: the caregiver has a Google account to sign in; the older adult needs no account.

3. Functional requirements

Format: As a <role>, I want <goal>, so that <benefit>. Every story traces to a pain from our M1 problem and interviews (P1–P4, 3–4 Sep 2026). Every acceptance criterion covers either the correct path or an error path and can be checked by a test.

FR-1 — Pairing

User story: As a signed-in caregiver, I want to pair my phone with my parent's phone by scanning a QR code or typing an invite code on my phone, so that I can set up medicines and receive alerts without my parent creating an account or typing anything.

Pain this traces to: older adults who are not comfortable with smartphones struggle with sign-up, passwords, and typing.

Security rules for this feature:

- The caregiver signs in with a Google account before pairing (NFR-9).
- The older adult has no account; on first open the app creates a hidden senior ID that the older adult never sees or types.
- The pairing code is random, works once, and expires after 10 minutes (NFR-7).
- Only caregivers linked to an older adult can see that person's data (NFR-5).

Acceptance criteria:

1. Given the older adult opens the app for the first time, when the start screen appears, then it shows a QR code and a 6-digit invite code and asks for no account, password, or typing.
2. Given a caregiver who is not signed in, when they try to pair, then they are asked to sign in first and no link is created.

3. Given a signed-in caregiver scans or enters a valid code, when pairing completes, then both phones show the pair as linked, with the caregiver's name on the older adult's screen.
4. Given an expired or already-used code, when the caregiver enters it, then pairing is rejected with a clear message and no link is created.
5. Given a linked caregiver, when the link is removed, then that caregiver immediately loses access to the older adult's data and is no longer listed on the older adult's screen.

FR-2 — Label scan → reviewed draft (AI-assisted)

User story: As a caregiver, I want to photograph a medication label and get a pre-filled draft of the medicine and its instructions, so that I don't have to type every medicine by hand.

Pain this traces to: setting up medicine names and times in an app takes many steps.

1. Given a clear label photo, when the caregiver submits it, then the system shows a draft (medicine name, amount per dose, time, instructions, and warnings) and saves nothing until the caregiver confirms.
2. Given the caregiver edits a field in the draft, when they confirm, then the edited value — not the AI's original guess — is saved.
3. Given a field the system cannot read, when the draft is shown, then that field is empty and marked "please check", never guessed.
4. Given a label that says to take only when needed (e.g. "ทานเฉพาะเมื่อมีอาการ"), when the draft is shown, then it is marked "as needed" and no fixed daily schedule is created automatically. (Added after our Gemini test missed this warning.)
5. Given the caregiver has set the older adult's meal times once (e.g. breakfast 07:30), when they confirm a medicine, then they choose which sessions it belongs to (e.g. after breakfast, after dinner), and each session's reminder time is taken from those meal times.

FR-3 — Session reminder

User story: As an older adult, I want a reminder at each medication time that shows all of that session's medicines on one screen, so that I take the right medicines without checking every bag.

Pain this traces to: older adults forget a dose or are unsure which medicines belong to which time.

1. Given a 08:00 session with two medicines, when it is 08:00, then a notification appears, and opening it shows both medicine names and photos on one screen.
2. Given notification permission is turned off, when setup finishes, then the app shows a clear message that reminders cannot appear and how to turn them on.

FR-4 — One-tap confirm

User story: As an older adult, I want to confirm the whole session with one tap, so that I don't have to confirm each pill and my family knows I took them.

Pain this traces to: caregivers cannot tell whether a dose was taken without calling.

3. Given an active reminder, when the older adult taps "Taken", then the session status becomes Confirmed with the time recorded, and no more reminders are sent for that session.
4. Given a session that is already confirmed, when "Taken" is tapped again, then no duplicate record is created.

FR-5 — Escalation to caregiver

User story: As a caregiver, I want to be told when a session is still not confirmed after repeated reminders, so that I only need to call when something may be wrong instead of checking every dose.

Pain this traces to: caregivers call to check because they have no other way to know the status of a dose.

1. Given a session with a limit of 2 re-reminders at 5-minute intervals (both configurable), when the older adult does not confirm, then they are reminded again at +5 and +10 minutes, and the caregiver is notified "not yet confirmed" after the last reminder.

- Given the older adult confirms after the first re-reminder, when the next reminder time arrives, then no reminder is sent and the caregiver is not notified.
- Given a caregiver alert, when it is shown, then it says "not yet confirmed" and never "missed" or "not taken".

4. Non-functional requirements

Forbidden words unless quantified: fast, easy, user-friendly, smart, secure. NFR-5 (security/privacy) is required for M2 — every SRS must have at least one.

The numbers below are our team's initial targets. We will confirm them with our interviewees and prototype tests and update this table if they change.

ID	Kind	Statement (testable)	How we will check
NFR-1	performance	A session reminder shall appear within 1 minute of its scheduled time.	Compare notification time with scheduled time on the test Android phone, 5 trials
NFR-2	performance	A label draft shall be shown within 10 seconds of submitting a photo on Wi-Fi.	Stopwatch, 5 labels
NFR-3	usability	At least 3 of 4 older-adult test users shall confirm a session within 10 seconds on their first try without help.	Timed trial with interviewees P1–P4 using the prototype
NFR-4	usability	The "Taken" button shall be at least 64×64 dp, and medicine names on the reminder screen at least 20 sp.	Measure in the prototype / design system
NFR-5	security / privacy (required)	A caregiver shall see only the older adults linked to them; a request for another older adult's data shall fail.	Change the older-adult ID in a request and confirm it is rejected
NFR-6	privacy	Notifications on the lock screen shall not show medicine names.	Lock the phone and inspect the notification
NFR-7	security	A pairing code shall expire after 10 minutes and work only once.	Reuse a code; wait 11 minutes and try again
NFR-8	reliability	A scheduled reminder shall still appear when the app is closed and after the phone restarts.	Close the app / restart the phone before a session and check the reminder arrives
NFR-9	security	A caregiver shall sign in with a Google account before viewing or changing any older adult's data; a signed-out caregiver shall see no senior data.	Sign out, then try to open a senior's page and confirm it asks to sign in.
NFR-10	security	All data between the phones and the server shall use encrypted connections (HTTPS); no plain HTTP.	Check every service URL in the code starts with https://
NFR-11	privacy	The system shall not store the older adult's national ID number, address, date of birth, phone number, or hospital number.	Review every field in the data model.

5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1 Pair devices	Caregiver, Older adult	FR-1	Caregiver scans the QR code on the older adult's phone; both phones show "linked".
UC-2 Add medicine from label	Caregiver	FR-2	Caregiver photographs a label, fixes one field in the draft, and confirms; the medicine and its sessions are saved.
UC-3 Receive session reminder	Scheduler → Older adult	FR-3	At 08:00 the older adult sees both morning medicines with photos on one screen.
UC-4 Confirm session	Older adult	FR-4	Older adult taps "Taken"; the session is confirmed and reminders stop.
UC-5 Escalate unconfirmed session	Scheduler → Caregiver	FR-5	No confirmation after two re-reminders; the caregiver gets "not yet confirmed" and calls.

5.2 Use-case diagram

Not drawn yet on purpose. Week 3's job is naming these five use cases correctly against the FR list above — the actor/association/<<include>> diagram itself is Week 4's task, taught in the next session.

6. Out of scope

1. Dosage advice, or changing or interpreting a prescription beyond what is printed on the label.
2. Reading handwritten labels.
3. Verifying that medicine was actually swallowed; "Taken" is self-reported.
4. Full-screen alarms that override the phone's normal notification settings.
5. An iOS version, payments, or integration with hospital systems.
6. Accounts or passwords for older adults, and storing their national ID, address, date of birth, phone number, or hospital number.

7. Traceability matrix (Golden Thread)

Only FR-3 is filled all the way across — that is the single thread this course follows week by week (Ch3 → Ch4/5 → Ch6 → Ch7). The other rows stay open until their week arrives, exactly like your own M2 should.

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Sign-up and typing are hard for older adults	FR-1 pairing	(Week 5)	(Week 6–7)	(Week 7)
Setting up medicines takes many steps	FR-2 label scan	(Week 5)	(Week 6–7)	(Week 7)
Older adults forget or are unsure which medicines to take	FR-3 reminder	(Week 5)	(Week 6–7)	(Week 7)
Caregivers cannot tell whether a dose was taken	FR-4 confirm	(Week 5)	(Week 6–7)	(Week 7)
Caregivers call to check every dose	FR-5 escalation	Session state machine: Scheduled → Reminded → Confirmed /	Caregiver "not yet confirmed" alert screen (planned)	FR-5 AC 1–3 scenarios (planned)

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
		Caregiver notified (planned)		

8. AI usage log

If this table is empty but the prose looks generated, the milestone is capped at 50% — log every prompt that drafted a story or paragraph, honestly.

Date	Tool model /	What we asked	What AI produced	What we changed / verified	What we learned
24 Aug 2026	Claude	Draft user stories and acceptance criteria for FR-1 to FR-5 from our M1 charter.	Five stories with the right roles (older adult, caregiver), but FR-2's draft listed "dose", FR-1 did not say the caregiver must be signed in, and no criterion said where a session's reminder time comes from.	Replaced "dose" with "amount per dose (e.g., 2 tablets)", because "ขนาดยา" can also mean strength (500 mg). Made FR-1 "a signed-in caregiver". Added FR-2 criterion 5: reminder times come from meal times the caregiver sets.	A story can read correctly and still be ambiguous. Each criterion must name the exact value the user sees (2 tablets, 07:30), or two people will build two different things.
24 Aug 2026	Gemini API	Read 2 de-identified medication labels and return the medicine details.	Correct name (Sinlumag), 2 tablets, and three meal times; silently dropped "ทานเฉพาะเมื่อนมีอาการ" (take only when needed).	Added "warnings" to the FR-2 draft fields and criterion 4: an as-needed label never creates a fixed schedule.	The AI only returns the fields we ask for. Safety text on a label needs its own field, or it disappears without any error.
24 Aug 2026	Claude	Draft non-functional requirements.	Measurable statements with numbers: reminder within 1 minute, draft within 10 seconds, 64 × 64 dp button, 10-minute pairing code.	Kept the numbers but labelled them team targets, to be confirmed with P1–P4 and prototype tests.	AI can make an NFR measurable, but it cannot know the right number for our users. Only our own tests can.
24 Aug 2026	Claude	Check whether FR-3's full-screen reminder that stays for 5 minutes is possible on Android and iOS.	Android 14+ limits full-screen alerts to alarm and calling apps, and an app cannot control how long a notification stays on screen.	Removed "full-screen" and "visible 5 minutes". FR-3 now uses a normal high-priority notification that opens a large reminder screen.	A requirement the operating system does not allow can never pass its own test. Platform limits must be checked before writing the criterion.

9. Self-check

- ✓ Every FR is a user story with a benefit — all 5 FRs in §3 end in a “so that” clause tied to a real pain.
- ✓ Every FR has acceptance criteria you could actually test — 2 AC per FR, each a number, a state, or a yes/no.
- ✓ No FR is vague — no bare “fast”, “easy”, or “user-friendly” anywhere in §3.

- ✓ NFRs exist, including security / privacy — NFR-5 is a required access-control check.
- ✓ Every FR traces back to a stated problem — see “Pain this traces to” under each FR, and §7.
- ✓ Scope and out-of-scope are stated — §2.1 and §6.
- ✓ Use-case list matches the FR list — 5 use cases, one per FR (diagram added Week 4).
- ✓ AI log is honest — §8 shows exactly what was asked, drafted, and hand-corrected.

Course rule: you may use AI to draft. You must log it, verify it against a real stakeholder, and be able to explain every FR in the oral exam. Work you cannot explain earns no credit.